Diffuse Cerebral and Cerebellar Atrophy–Intractable Seizures–Progressive Microcephaly Syndrome (QARS1-Related Disorder)

Disease: Diffuse Cerebral and Cerebellar Atrophy–Intractable Seizures–Progressive Microcephaly Syndrome (MSCCA) **MONDO ID:** MONDO:0014335 · **OMIM:** 615760 · **Causal gene: QARS1** (glutaminyl-tRNA synthetase 1; HGNC:9751; formerly *QARS*)

Summary

Diffuse Cerebral and Cerebellar Atrophy–Intractable Seizures–Progressive Microcephaly Syndrome (MSCCA; MONDO:0014335, OMIM 615760) is an ultra-rare autosomal-recessive neurodevelopmental and neurodegenerative disorder caused by biallelic (compound-heterozygous or homozygous) hypomorphic loss-of-function variants in *QARS1*, the gene encoding cytosolic glutaminyl-tRNA synthetase (GlnRS; EC 6.1.1.18). GlnRS is a ubiquitously expressed housekeeping enzyme that charges tRNA^Gln with glutamine, an obligatory step in cytoplasmic protein synthesis. The original disease-defining report identified compound-heterozygous *QARS1* mutations as the only candidate causative variants in two unrelated families presenting with progressive microcephaly, severe infantile-onset seizures, and atrophy of the cerebral cortex and cerebellar vermis (PMID: 24656866).

The core mechanism is **reduced aminoacylation capacity combined with protein destabilization/misfolding**. Variants in the N-terminal domain (NTD) reduce catalytic activity >10-fold and decrease enzyme stability, whereas variants in the catalytic core (e.g., p.Arg403Trp, p.Arg515Trp) abolish aminoacylation entirely; truncating alleles (e.g., p.Lys496*) produce diminished, aggregation-prone protein (PMID: 24656866; PMID: 26869582; PMID: 25471517). The resulting deficit in glutaminyl-tRNA^Gln impairs translation in the developing

brain, which appears exquisitely dependent on QARS1, producing progressive postnatal microcephaly, intractable seizures, and cerebral/cerebellar atrophy with profound intellectual disability.

The phenotype is a **severity continuum governed by residual enzyme activity.** At the severe end are neonatal-onset epileptic encephalopathies with marked brain atrophy; at the milder end are patients with non-epileptic microcephaly/intellectual disability or late-childhood focal epilepsy with good seizure control. Management is entirely symptomatic — antiseizure medication, developmental support, nutrition — with no disease-modifying therapy. A mechanistically plausible but QARS1-unproven downstream effector linking aminoacyl-tRNA synthetase (ARS) deficiency to seizures and developmental delay is the **integrated stress response (ISR)**, demonstrated for other ARS disorders.

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1. Disease Information

Overview. MSCCA is a Mendelian, autosomal-recessive disorder of cytoplasmic translation. Affected children typically present in the neonatal period or early infancy with intractable seizures, show progressive (postnatal) microcephaly, and develop diffuse cerebral and cerebellar atrophy on neuroimaging, accompanied by severe-to-profound global developmental delay/intellectual disability. The disorder was defined molecularly in 2014 (PMID: 24656866).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0014335
OMIM	615760
Causal gene	QARS1 (HGNC:9751), chromosome 3p21.31
Gene OMIM	603727
Enzyme	Glutaminyl-tRNA synthetase, EC 6.1.1.18

Synonyms / alternative names. Microcephaly, progressive, seizures, and cerebral and cerebellar atrophy (MSCCA); QARS1-related (glutaminyl-tRNA synthetase) deficiency syndrome; QARS1-associated developmental and epileptic encephalopathy (DEE); QARS1 encephalopathy.

Information source. Disease-level knowledge is aggregated from OMIM/Orphanet and from a modest number of primary case reports and small case series (individual-patient data). There is no EHR-scale cohort; the evidence base is small-N clinical-genetics literature (~26 published patients as of the most recent review, PMID: 40448856).

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2. Etiology

Primary cause — genetic. MSCCA is caused exclusively by **biallelic loss-of-function variants in *QARS1***. There is no environmental or infectious cause of the disease itself. Both compound-heterozygous and homozygous (often consanguineous) genotypes are reported.

Genetic risk factors. The only established risk factor is inheritance of two pathogenic *QARS1* alleles. **Consanguinity** increases risk of homozygous presentation (e.g., the Pakistani family homozygous for c.1133G>A p.(Arg378His), PMID: 40448856). No modifier genes have been formally identified; residual enzyme activity of the specific allele combination is the principal determinant of severity (functional, not a separate locus).

Environmental risk / protective factors. None established. As a monogenic translational disorder, there are no known dietary, lifestyle, occupational, or toxic risk or protective factors. One notable **gene–environment interaction at the phenotype level**: intercurrent infection can trigger acute para-infectious encephalopathy in milder QARS1 patients (steroid-responsive), suggesting reduced translational reserve lowers the threshold for metabolic decompensation during illness (PMID: 39715963).

Protective factors. The only "protective" factor is retained residual QARS1 catalytic activity/solubility, which attenuates severity (e.g., variants with conserved aminoacylation but reduced solubility give milder disease, PMID: 34774383).

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3. Phenotypes

Phenotypes are drawn from OMIM, published case reports, and series. Frequencies are approximate given the small cohort.

Phenotype	Type	HPO term	Onset	Frequency	Severity / course
Progressive microcephaly	Physical/ sign	HP:0000253	Postnatal (often normal OFC at birth)	Near-universal	Severe, progressive
Seizures / epilepsy	Sign	HP:0001250	Neonatal–infancy (variable)	~86% (drug-resistant in ~79%)	Severe; often intractable
Developmental and epileptic encephalopathy	Clinical syndrome	HP:0200134	Infancy	Common	Severe
Cerebral cortical atrophy	Imaging/ sign	HP:0002120	Infancy, progressive	Common	Progressive
Cerebellar atrophy (vermis > hemispheres)	Imaging/ sign	HP:0001272	Infancy, progressive	Common	Progressive
Severe/profound intellectual disability	Cognitive	HP:0010864	From infancy	Near-universal	Severe
Global developmental delay	Behavioral	HP:0001263	Infancy	Near-universal	Severe
Spasticity / hypertonia	Sign	HP:0001257	Infancy–childhood	Frequent	Variable
Muscular hypotonia (axial)	Sign	HP:0001252	Infancy	Frequent	Variable
Feeding difficulties / failure to thrive	Sign	HP:0011968 / HP:0001508	Infancy	Frequent	Variable
Short stature / growth failure	Sign	HP:0004322	Postnatal	Reported subset	Severe in some

Phenotype	Type	HPO term	Onset	Frequency	Severity / course
Movement disorder (orofacial dyskinesia, stereotypies)	Sign	HP:0002072	Infancy	Reported	Variable

Age of onset. Typically neonatal to early infantile for seizures; microcephaly is usually postnatal and progressive. Milder cases show later childhood-onset focal epilepsy (e.g., 3.5 years, PMID: 39715963).

Phenotypic spectrum (key finding). The disorder spans a continuum: - **Severe end:** neonatal/infantile drug-resistant epileptic encephalopathy with marked cerebral–cerebellar atrophy (classic, PMID: 24656866). - **Intermediate:** severe growth failure, microcephaly, characteristic facies, intellectual disability, with *absent* epilepsy and only minor MRI changes — "The clinical presentation of our patients differs from the original QARS-associated syndrome in the severe postnatal growth failure, absence of epilepsy, and minor MRI findings, thus further expanding the phenotypic spectrum" (PMID: 28620870). - **Non-epileptic end:** microcephaly and severe intellectual/developmental disability without epilepsy — "The phenotype in the affected individuals of the family was characterized by microcephaly, severe intellectual and developmental disability, but no epilepsy" (PMID: 40448856). - **Mild end:** controllable focal epilepsy (carbamazepine, levetiracetam) with minimal/absent cerebellar atrophy (PMID: 34774383; PMID: 39715963).

Quality-of-life impact. Severe cases have profound impact: non-ambulatory status, no expressive language, dependence for all activities of daily living, and medically refractory epilepsy. No formal EQ-5D/SF-36/PROMIS data exist for this ultra-rare disease.

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4. Genetic / Molecular Information

Causal gene. *QARS1* (glutaminyl-tRNA synthetase 1), HGNC:9751, chromosome 3p21.31; encodes a cytosolic class I aminoacyl-tRNA synthetase. Reference transcript NM_005051.2.

Pathogenic variants (representative).

Variant (protein)	Domain	Functional effect	Source
p.Gly45Val	N-terminal domain	>10-fold reduced aminoacylation	PMID: 24656866
p.Tyr57His	N-terminal domain	>10-fold reduced aminoacylation	PMID: 24656866
p.Arg403Trp	Catalytic core	Aminoacylation abolished	PMID: 24656866
p.Arg515Trp	Catalytic core	Aminoacylation abolished	PMID: 24656866
p.Lys496* (truncating)	Catalytic domain	Diminished, aggregated protein (LoF)	PMID: 25471517
p.Arg378His (homozygous)	—	Non-epileptic phenotype	PMID: 40448856
p.Arg378Cys + p.Arg525Gln	—	Reduced solubility; mild ~2-fold ↓ efficiency; mild phenotype	PMID: 34774383
p.Val476Ile (homozygous)	—	Growth failure, microcephaly, no epilepsy	PMID: 28620870
p.Tyr435Cys + p.Arg267Trp	—	Mild focal epilepsy; para-infectious encephalopathy	PMID: 39715963

Variant classification. Pathogenic / likely pathogenic per ACMG/AMP; several were initially VUS and later upgraded after functional testing (PMID: 34774383). **Variant types:** predominantly missense, plus nonsense/truncating alleles. The original report emphasized that "The four QARS mutations altered highly conserved amino acids, and the aminoacylation activity of QARS was significantly impaired in mutant cell lines" (PMID: 24656866).

Allele frequency. Pathogenic alleles are extremely rare in gnomAD, consistent with recessive, severe disease. **Origin:** germline. **Functional consequence:** loss of function (reduced catalytic efficiency and/or protein destabilization/aggregation); no gain-of-function or dominant-negative mechanism.

Modifier genes. None established. Severity tracks with residual enzyme activity of the allele combination (genotype–phenotype correlation).

Epigenetic / chromosomal abnormalities. None implicated; this is a single-gene point-mutation disorder, not a copy-number/structural disorder.

CHEBI entities relevant to the enzyme: L-glutamine (CHEBI:18050), ATP (CHEBI:15422), aminoacyl-tRNA/Gln-tRNA^Gln (CHEBI:78527 family), AMP (CHEBI:16027), diphosphate/pyrophosphate (CHEBI:33019), Mg^{2+} (CHEBI:18420).

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5. Environmental Information

Environmental factors: None causal. **Lifestyle factors:** Not applicable (onset in infancy; monogenic). **Infectious agents:** No infectious cause. However, intercurrent infection can precipitate acute para-infectious (steroid-responsive) encephalopathic decompensation in milder QARS1 patients, indicating reduced translational reserve under metabolic stress (PMID: 39715963).

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6. Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic hypomorphic *QARS1* variants** (NTD or catalytic-core missense, or truncating) → **reduced or absent GlnRS catalytic activity and/or destabilized, misfolded, aggregation-prone enzyme** (demonstrated in vitro; PMID: 24656866; PMID: 26869582; PMID: 25471517).
2. Loss of GlnRS function **leads to** reduced charging of tRNA^Gln with glutamine → **diminished supply of glutaminyl-tRNA^Gln** for the ribosome.
3. Reduced aminoacylated tRNA^Gln **results in** impaired/slowed cytoplasmic protein synthesis, with the **developing brain being exceptionally sensitive** (high translational demand of neurogenesis and synaptogenesis) — *inferred* from tissue-selectivity of the phenotype.
4. **Branch A (translational insufficiency):** impaired translation **leads to** failure of normal brain growth and neuronal survival → **progressive postnatal microcephaly and cerebral/cerebellar atrophy** (clinically demonstrated).
5. **Branch B (integrated stress response — inferred):** undercharged tRNA and translational stress **plausibly activate the ISR** (GCN2→eIF2α-P→ATF4). Across ARS disorders ISR

activation causally contributes to disease, and in a *PARS2* Drosophila model reproduces developmental delay and seizures (PMID: 34516839; PMID: 41380592). *No QARS1-specific ISR demonstration exists — this step is inferred.*

6. Branches A + B **converge on** cortical neuronal dysfunction and loss → **intractable seizures, severe intellectual disability, and ongoing neurodegeneration** (clinically demonstrated).
7. **Modulating factor:** residual catalytic activity/solubility of the allele combination **determines** position along the severity continuum (neonatal DEE ↔ non-epileptic microcephaly).

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QARS1 biallelic LoF variants
        │
        ▼
↓ GlnRS activity / misfolded enzyme ──► ↓ Gln-tRNA^Gln charging
        │
        ▼
Impaired brain protein synthesis
   ├── Branch A: failed neuronal growth/survival ──► microcephaly, cerebral/cerebellar atrophy
   └── Branch B (inferred): ISR activation (GCN2–eIF2α–ATF4) ──► developmental delay, seizures
        │
        ▼
Cortical neuronal dysfunction & loss ──► intractable seizures + severe ID
        │
   (residual activity sets severity: DEE ◄──► non-epileptic microcephaly)
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Supporting detail

Enzymology (upstream lesion). GlnRS is a **class I aminoacyl-tRNA synthetase** with a mechanistically unusual requirement: it **cannot activate glutamine with ATP until tRNA^Gln is bound** (tRNA-dependent amino-acid activation). "Glutaminyl-tRNA synthetase generates Gln-tRNA(Gln) 10(7)-fold more efficiently than Glu-tRNA(Gln) and requires tRNA to synthesize the activated aminoacyl adenylate in the first step of the reaction" (PMID: 15845536). tRNA-mediated induced fit assembles the active site (PMID: 16734422). Amino-acid discrimination uses a negative determinant — "Arg30 functions as a negative determinant to drive binding of non-cognate glutamate into a non-productive orientation" (PMID: 12691748) — and Tyr211 hydrogen bonding (PMID: 9562563). This intricate active-site architecture explains why catalytic-core substitutions (Arg403, Arg515) are catastrophic.

Protein dysfunction (two mechanistic classes). Crystal structures of human GlnRS show that "Pathological mutations mapping in the NTD alter the domain structure, and decrease catalytic activity and stability of GlnRS, whereas missense mutations in the catalytic domain induce misfolding of the enzyme" (PMID: 26869582). Truncating p.Lys496 *shows "diminished and aberrantly aggregated expression, indicating the loss-of-function nature of this mutant" (PMID: 25471517). Milder alleles reduce solubility** more than catalytic efficiency (PMID: 34774383).

Molecular pathways / cellular processes. Core pathway: **aminoacyl-tRNA biosynthesis → cytoplasmic translation (GO:0006412)**; GlnRS is a component of the cytoplasmic multi-tRNA synthetase complex (MSC), assembled via leucine-zipper motifs with RARS1/AIMP1/AIMP2 (PMID: 39542129). Downstream (inferred): **integrated stress response / eIF2α signaling** and neuronal apoptosis. ARSs also have moonlighting roles (transcription, apoptosis, angiogenesis, inflammation; PMID: 37776328), raising the possibility that loss of a non-canonical QARS1 function contributes, though this is unproven for MSCCA.

Immune involvement. No primary autoimmune mechanism. Notably, in a related ARS disorder (mt-AspRS/DARS2 neuronal knockout), immune/stress pathways are upregulated *before* overt atrophy (PMID: 31887305) — a possible parallel, not demonstrated for QARS1.

Suggested ontology terms. GO:0006425 (glutaminyl-tRNA aminoacylation), GO:0006412 (translation), GO:0004819 (glutamine-tRNA ligase activity), GO:0034198 / amino-acid-starvation response (ISR), GO:0006915 (apoptotic process); CL:0000540 (neuron), CL:0000121 (Purkinje cell), CL:0000127 (astrocyte).

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7. Anatomical Structures Affected

Organ level. Primary: **brain** (central nervous system). The disorder is essentially neurological; other organs are largely spared, though growth failure/short stature occurs in a subset (PMID: 28620870).

Regional localization (bilateral, diffuse). - **Cerebral cortex** (UBERON:0000956) — diffuse atrophy (HP:0002120). - **Cerebellum**, especially **vermis** (UBERON:0004720 / UBERON:0002037) with milder hemispheric atrophy — the defining report described "atrophy of the cerebral cortex and cerebellar vermis, and mild atrophy of the cerebellar hemispheres" (PMID: 24656866). - White matter: variable; acute T2 hyperintensities during para-infectious episodes (PMID: 39715963).

Tissue / cell level. Nervous tissue; affected populations include cortical **neurons** (CL:0000540) and cerebellar neurons including **Purkinje cells** (CL:0000121). QARS1 is ubiquitously expressed; selective CNS vulnerability reflects high translational demand during brain development.

Subcellular level. Cytoplasm / cytosol (GO:0005829) — site of cytoplasmic aminoacylation and the multi-tRNA synthetase complex. (QARS1 is cytosolic, distinct from mitochondrial ARSs.)

Lateralization. Bilateral and diffuse/symmetric.

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8. Temporal Development

Onset. Congenital-to-early-infantile. Seizures often begin in the neonatal period or first months; microcephaly is typically **postnatal and progressive** (head circumference may be normal at birth). Onset pattern is chronic/insidious for atrophy, with acute seizure presentation.

Progression. Progressive neurodegeneration: microcephaly worsens, and cerebral/cerebellar atrophy advances over infancy and childhood. Course is generally **progressive** in classic cases; milder cases may be relatively more static with controllable epilepsy (PMID: 34774383).

Patterns. Seizures are frequently relapsing/intractable; some milder cases achieve sustained seizure freedom on monotherapy (PMID: 39715963). **Critical period:** brain development in infancy — the window of greatest vulnerability and, theoretically, greatest potential benefit from any future intervention. Episodic **para-infectious encephalopathic decompensations** occur in some milder patients (PMID: 39715963).

Duration. Chronic, lifelong.

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9. Inheritance and Population

Epidemiology. Ultra-rare; prevalence/incidence not formally estimated. ~26 patients published to date (PMID: 40448856). No reliable per-100,000 figures exist.

Inheritance. Autosomal recessive. Both compound-heterozygous and homozygous genotypes reported.

Penetrance / expressivity. Penetrance appears complete for biallelic loss-of-function genotypes; **expressivity is highly variable** (severe DEE to non-epileptic microcephaly), driven by residual enzyme activity.

Genetic anticipation: Not applicable (not a repeat-expansion disorder). **Germline mosaicism:** not specifically reported. **Consanguinity:** important for homozygous presentations (e.g., consanguineous families; PMID: 40448856; PMID: 28620870). **Founder effects:** none well established; individual recurrent variants noted in specific families/populations. **Carrier frequency:** not established (expected very low given rarity).

Population demographics. Cases reported across multiple populations (Turkish, Pakistani, Ashkenazi-Jewish, East/Southeast Asian, European). No strong sex bias reported. Age distribution: predominantly pediatric given early onset and severe course.

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10. Diagnostics

Genetic testing (definitive). Diagnosis rests on identification of **biallelic pathogenic *QARS1* variants**, typically via **trio whole-exome sequencing (WES)** or whole-genome sequencing; targeted epilepsy/neurodevelopmental gene panels including QARS1 are also used. Single-gene testing is appropriate when the phenotype is recognized. **Functional validation** (aminoacylation assays, solubility) has been pivotal to reclassify VUS to pathogenic (PMID: 34774383; PMID: 24656866). Chromosomal microarray/karyotype/FISH are generally uninformative (point-mutation disorder). Mitochondrial DNA testing is not indicated (QARS1 is cytosolic).

Imaging. Brain MRI is central: progressive microcephaly with **cerebral cortical atrophy and cerebellar (vermian-predominant) atrophy** (PMID: 24656866). Milder cases may show only minor volume loss (PMID: 34774383; PMID: 28620870). Para-infectious episodes may show reversible white-matter T2 hyperintensity with diffusion restriction (PMID: 39715963).

Electrophysiology. EEG shows epileptiform abnormalities (e.g., shifting/multifocal foci, occipital spikes); useful for seizure characterization but not specific (PMID: 34774383; PMID: 39715963).

Laboratory / biomarkers. No specific blood/urine biomarker. Diagnosis is molecular; aminoacylation/solubility assays are research-grade functional confirmations. No validated omics diagnostic.

Clinical criteria / differential diagnosis. Diagnosis is genetic; the clinical gestalt (progressive microcephaly + intractable infantile seizures + cerebral–cerebellar atrophy + severe ID) prompts testing. Differential diagnosis includes other **ARS-related encephalopathies** (mitochondrial and cytosolic ARS disorders), congenital/postnatal microcephaly syndromes, pontocerebellar hypoplasias, and other genetic developmental and epileptic encephalopathies; distinguished by gene-specific testing.

Screening. No population newborn screening. Carrier/cascade testing within families and prenatal/preimplantation testing are feasible once the familial variants are known.

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11. Outcome / Prognosis

Severity and mortality. Classic severe disease carries a poor prognosis: profound disability, intractable epilepsy, and risk of early death; precise survival statistics are unavailable for this ultra-rare condition. Milder genotypes have more favorable outcomes with controlled epilepsy and longer survival.

Morbidity / function. Severe cases are non-ambulatory, nonverbal, and fully dependent, with refractory seizures. Milder cases achieve seizure control and variable developmental progress (PMID: 34774383; PMID: 39715963). No standardized QoL instruments have been applied.

Complications. Status epilepticus, aspiration/feeding complications, failure to thrive, and infection-triggered encephalopathic decompensations (PMID: 39715963).

Prognostic factors. The strongest predictor is **residual QARS1 enzyme activity/solubility**: variants retaining function (reduced solubility but near-normal catalysis) predict milder phenotypes (PMID: 34774383); catalytic-null alleles predict severe disease (PMID: 24656866).

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12. Treatment

No disease-modifying therapy exists. Management is **symptomatic and supportive.**

Pharmacotherapy — antiseizure medications (NCIT: Anticonvulsant Agent / Antiepileptic Agent). Seizures are often drug-resistant (~79% pharmaco-resistant historically; PMID: 34774383). Nonetheless, some patients respond well: - **Carbamazepine** achieved seizure control in a milder case (PMID: 34774383). - **Levetiracetam** monotherapy produced a year of

seizure freedom in another milder patient (PMID: 39715963). - **Intravenous methylprednisolone / corticosteroids** for acute para-infectious encephalopathy episodes, with optimal response reported (PMID: 39715963).

Supportive / rehabilitative. Developmental therapies (physical, occupational, speech), nutritional support/feeding management, management of spasticity and tone, and multidisciplinary neurodevelopmental care.

Advanced / experimental therapeutics. None approved. Conceptually, loss-of-function translational disorders are candidates for future strategies (gene replacement, allele-specific solubility rescue for misfolding alleles, ISR modulation), but **no gene, cell, or RNA therapy and no clinical trials** exist for QARS1 disease.

Pharmacogenomics / personalized medicine. Limited to genotype-informed prognostication (residual-activity–based severity prediction).

Suggested NCIT terms: Carbamazepine (C341), Levetiracetam (C29295), Methylprednisolone (C636), Anticonvulsant Agent (C264), Supportive Care (C45984).

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13. Prevention

Primary prevention. Not applicable to the biological cause (monogenic). The principal preventive levers are **reproductive/genetic**: genetic counseling for at-risk (often consanguineous) families, carrier testing, cascade testing, and prenatal or preimplantation genetic testing once familial *QARS1* variants are known.

Secondary/tertiary prevention. Early genetic diagnosis enables anticipatory management, avoidance of unnecessary investigations, and vigilance for infection-triggered decompensation; prompt treatment of intercurrent illness and consideration of steroids for para-infectious encephalopathy (PMID: 39715963). Early seizure management and developmental support constitute tertiary prevention.

Counseling. Autosomal-recessive recurrence risk is 25% per pregnancy for carrier couples; genetic counseling is central.

Immunization / public health / environmental interventions. Not applicable to disease causation (standard childhood immunization and infection-avoidance are prudent given decompensation risk).

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14. Other Species / Natural Disease

Taxonomy / orthologs. QARS1 is evolutionarily conserved; orthologs exist across eukaryotes (human *QARS1*; mouse *Qars1*). Mechanistic enzymology derives substantially from bacterial/archaeal and fungal GlnRS systems (*E. coli*, *Thermus thermophilus*, *Pseudomonas aeruginosa*, *Candida albicans* Gln4) (PMID: 16734422; PMID: 30248426; PMID: 31833153; PMID: 38402621).

Natural disease in other species. No naturally occurring QARS1-equivalent disease is well documented in companion animals or wildlife (no established OMIA counterpart). **Comparative biology:** the aminoacylation mechanism is deeply conserved, making bacterial/fungal GlnRS structures directly informative about human pathogenic residues. **Zoonotic/transmission:** not applicable.

Note: Fungal GlnRS (Gln4) is pursued as an *antifungal drug target* — a translational-biology counterpoint, not a disease model (PMID: 38402621).

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15. Model Organisms

Direct QARS1 disease models are limited. No well-characterized mouse knock-in model recapitulating MSCCA is reported in this knowledge base. **In vitro / cellular models** have been central: - **Patient-derived cell lines / transfected cells:** aminoacylation and stability assays demonstrating reduced catalytic activity and misfolding (PMID: 24656866). - **Neuroblastoma 2A (N2a) cells:** transient expression of p.Lys496 *showed diminished, aggregated protein (PMID: 25471517). - Recombinant human GlnRS crystallography:** structural basis of NTD vs catalytic-domain mutation effects (PMID: 26869582).

Analogous ARS models (supporting the mechanistic framework, not QARS1-specific). - ***Drosophila* PARS2 (mitochondrial prolyl-tRNA synthetase) model:** "Inactivation of dPARS2 in Drosophila causes developmental delay and seizure, two main clinical features in human PARS2 deficiency-associated patients," via ISR activation (PMID: 41380592). - **Neuronal *Dars2* knockout mice:** progressive cortical atrophy, behavioral disruption, and early immune/stress-pathway activation (PMID: 31887305).

Model limitations. Bacterial/fungal and in-vitro systems capture enzymology but not CNS-selective neurodegeneration; cross-ARS animal models demonstrate ISR-linked seizures/delay but are not QARS1. A faithful QARS1 hypomorphic mammalian model remains a gap.

Key Findings (expanded)

F001 — Biallelic QARS1 loss-of-function variants cause MSCCA

Whole-exome sequencing identified compound-heterozygous *QARS1* mutations as the only candidate causative variants in two unrelated families with progressive microcephaly, infantile seizures, and cerebral–cerebellar atrophy. "we report the identification of mutations in QARS (encoding glutaminyl-tRNA synthetase [QARS]) as the causative variants in two unrelated families affected by progressive microcephaly, severe seizures in infancy, atrophy of the cerebral cortex and cerebellar vermis, and mild atrophy of the cerebellar hemispheres" (PMID: 24656866). Four variants (NTD: p.Gly45Val, p.Tyr57His; catalytic core: p.Arg403Trp, p.Arg515Trp) altered conserved residues and impaired aminoacylation — NTD variants >10-fold, catalytic-core variants completely — establishing loss of function.

F002 — Dual mechanism: reduced catalytic efficiency plus misfolding

Human GlnRS crystal structures revealed two mechanistic classes: "Pathological mutations mapping in the NTD alter the domain structure, and decrease catalytic activity and stability of GlnRS, whereas missense mutations in the catalytic domain induce misfolding of the enzyme" (PMID: 26869582). A truncating allele behaves as a protein-destabilizing LoF: p.Lys496* produced "diminished and aberrantly aggregated expression" in N2a cells (PMID: 25471517).

F003 — Broad phenotypic spectrum from severe DEE to non-epileptic microcephaly

Severity tracks residual enzyme function. Severe neonatal epileptic encephalopathy anchors one end (PMID: 24656866); a consanguineous family of seven with homozygous p.(Arg378His) had "microcephaly, severe intellectual and developmental disability, but no epilepsy" (PMID: 40448856); and a homozygous p.Val476Ile presentation showed "severe postnatal growth failure, absence of epilepsy, and minor MRI findings, thus further expanding the phenotypic spectrum" (PMID: 28620870). Milder compound-heterozygous cases achieve seizure control (PMID: 34774383; PMID: 39715963).

F004 — Integrated stress response as a plausible shared effector (inferred)

Across ARS disorders, reduced aminoacylation activates the ISR, which causally contributes to pathology. In a Drosophila PARS2 model, "Inactivation of dPARS2 in Drosophila causes developmental delay and seizure, two main clinical features in human PARS2 deficiency-

associated patients" via ISR (PMID: 41380592); ISR also drives ARS-associated peripheral neuropathy (PMID: 34516839). No QARS1-specific ISR demonstration exists — this remains inferred.

F005 — GlnRS is a class I synthetase with unusual tRNA-dependent glutamine activation

GlnRS uniquely requires prior tRNA binding to activate glutamine: it "generates Gln-tRNA(Gln) 10(7)-fold more efficiently than Glu-tRNA(Gln) and requires tRNA to synthesize the activated aminoacyl adenylate in the first step of the reaction" (PMID: 15845536). Cognate selection uses "Arg30 ... as a negative determinant to drive binding of non-cognate glutamate into a non-productive orientation" (PMID: 12691748) and Tyr211 hydrogen bonding (PMID: 9562563). This explains why catalytic-core substitutions abolish activity.

Mechanistic Model / Interpretation (synthesis)

MSCCA is best understood as a **translational-insufficiency disorder of the developing brain.** Two mutation classes converge on loss of GlnRS function: (i) NTD missense variants that modestly reduce catalysis and destabilize the enzyme, and (ii) catalytic-core missense/truncating variants that abolish activity or cause aggregation. The shared endpoint is **insufficient glutaminyl-tRNA^Gln**, which throttles protein synthesis when the brain's biosynthetic demand peaks. This produces the triad of progressive microcephaly, cerebral/cerebellar atrophy, and intractable seizures. The **severity continuum maps onto residual activity/solubility**, explaining why some biallelic patients have neonatal epileptic encephalopathy while others have non-epileptic microcephaly or well-controlled focal epilepsy. A biologically compelling but **QARS1-unproven amplifier** is the **integrated stress response**, which in other ARS disorders is sufficient to generate seizures and developmental delay in vivo.

Mutation class	Example	Catalysis	Stability/ solubility	Typical severity
NTD missense	p.Gly45Val, p.Tyr57His	↓ >10-fold	↓	Severe
Catalytic-core missense	p.Arg403Trp, p.Arg515Trp	Abolished	—	Severe
Truncating	p.Lys496*	Lost	Aggregated	Severe
Solubility-predominant	p.Arg378Cys/ p.Arg525Gln	~2-fold ↓	↓↓ solubility	Mild

Evidence Base

PMID	Role	Key support
24656866	Landmark	Establishes QARS1 as causal; defines NTD vs catalytic-core variant effects
26869582	Mechanism	Human GlnRS structures: NTD → ↓activity/stability; catalytic domain → misfolding
25471517	Mechanism	p.Lys496* truncation → aggregated, diminished protein (LoF)
40448856	Spectrum	Non-epileptic phenotype; 26-patient review
28620870	Spectrum	Growth failure/microcephaly without epilepsy; phenotypic expansion
34774383	Spectrum/ therapy	Mild phenotype from solubility-reducing variants; 86% epilepsy/ 79% drug-resistant; carbamazepine control
39715963	Spectrum/ therapy	Childhood focal epilepsy; levetiracetam control; steroid-responsive para-infectious encephalopathy
15845536	Enzymology	tRNA-dependent amino-acid activation; 10^7-fold Gln selectivity
16734422	Enzymology	tRNA-mediated induced-fit active-site assembly
12691748	Enzymology	Arg30 negative determinant for amino-acid discrimination
9562563	Enzymology	Tyr211 hydrogen-bond recognition of glutamine
41380592	Inferred mechanism	PARS2 ISR model reproduces delay + seizures
34516839	Inferred mechanism	ISR contributes to ARS-associated neuropathy
31887305	Analogous model	DARS2 neuronal KO: progressive atrophy + early immune/stress activation
39542129	Context	QARS1 within multi-tRNA synthetase complex
37776328	Context	ARS moonlighting functions (apoptosis, transcription)
34774383	Epidemiology	86% of prior QARS1 patients had epilepsy; 79% pharmaco-resistant

Limitations and Knowledge Gaps

- **Small evidence base:** ~26 published patients; frequencies and prognosis are approximate, and no registry/natural-history study exists.
- **ISR step is inferred, not demonstrated for QARS1:** the integrated stress-response link rests on analogy to PARS2/MARS1/other ARS disorders, not direct QARS1 experiments.
- **No faithful mammalian QARS1 disease model** to establish the causal chain in vivo or test therapies.
- **Genotype–phenotype correlation is qualitative:** residual-activity thresholds that separate epileptic from non-epileptic phenotypes are not quantitatively defined.
- **No biomarkers, no disease-modifying therapy, no clinical trials.**
- **Moonlighting/apoptosis contribution unquantified** for QARS1 specifically.

Proposed Follow-up Experiments / Actions

1. **Generate a Qars1 hypomorphic mouse** (knock-in of p.Arg403Trp or an NTD variant) to test whether reduced aminoacylation produces progressive microcephaly, cerebellar atrophy, and seizures, and to map the causal chain in vivo.
2. **Directly assay ISR activation** (eIF2α-P, ATF4, CHOP) in QARS1-mutant patient iPSC-derived neurons/organoids, and test whether ISR inhibitors (e.g., ISRIB) rescue translational and survival phenotypes — converting the inferred Branch B into a tested step.
3. **Quantitative genotype–function–phenotype mapping:** systematically measure aminoacylation kinetics and solubility for all reported variants and correlate with clinical severity to define a predictive threshold for epilepsy risk.
4. **Cerebral organoid modeling** to capture CNS-selective vulnerability and screen mechanism-based therapeutics (ISR modulation, chemical chaperones for misfolding alleles).
5. **Establish an international patient registry** for reliable natural-history, survival, seizure-response, and QoL data, with standardized MRI/EEG phenotyping.
6. **Prospectively evaluate corticosteroid protocols** for para-infectious encephalopathic decompensations, given anecdotal benefit.

Evidence source legend: human clinical (case reports/series), in vitro (aminoacylation/solubility/structural), model organism (Drosophila/mouse for analogous ARS genes), computational/structural (crystallography, docking).
